Classification with Scholastic Travel Company

Team Members (GROUP 23): Gavin Loyd , Cierra Oliveira , Avery Trinidad
Date: February 26, 2026
Course: 90803-B: Machine Learning Foundations with Python

Executive Summary

Objective: Scholastic Travel Company (STC) is a multi-million dollar educational tourism firm. This project seeks to identify B2B clients that are likely to renew for services in the 2013-2014 school year, such that key segments are better retained and overall expenditure is minimized. We thus use multiple machine learning models to identify the most important contributing factors to customer retention, including trip cadence, novelty of account, and individual school profiles.

Methodology: We implement, in-succession, K-Nearest Neighbors, XGBoost (a tree-based model), and Support Vector Machines to categorize customers as either predicted to be retained (1) or otherwise to "churn" (0). L1 Regularization is used in order to identify those features most important for determining next-year account renewal. We also perform various optimization methods in order to maximize the accuracy and efficacy of our deployed models. From this, we produce a series of behavioral predictions for customers for the 2013-2014 term.

Recommendations: Given our findings, we suggest the following tailored strategic initiatives:

1. **Marketing streams and account management should be separated by "annual" status:** STC should handle non-annual trips as an entirely different product vertical.
2. **Retention efforts should heavily target brand new clients**: schools that were new to using STC services were distinctly more likely to churn compared to existing peers.
3. **"Single grade" trips can be deprioritized in STC's marketing strategy**: this reliable customer base should not exist in the same marketing stream as volatile new clients.
4. **Marketing efforts should emphasize STC's capacity to execute heavily attended, multi-day trips:** the more attendees of a given trip, the more likely it would be renewed.

Baseline Pre-Processing and Feature Engineering

Before model training, the dataset was accessed through a public DropBox link. "Retained" and "Churned" categories were binarily encoded as "1" and "0" respectively for later processing. All date-related fields had their months and years extracted, so as to measure possible seasonality, and were later converted into UNIX epoch day equivalents. (This measures their distance, in whole days, from January 1, 1970.)

The data was otherwise unaltered, as no programmatic null values were contained within the set. Select fields contain "NA" string values, which instead express that conditions for a particular column were "non-applicable" or otherwise not relevant for a particular record.

K-Nearest Neighbors

Initial Model

As a baseline, we first built a naive K-Nearest Neighbors model. As instructed, this initial model used the following features and parameters:

- *Predictive Features (4)*: ToGrade, Tuition, Days, TotalSchoolEnrollment
- *Parameters*: Number of Neighbors (K): 20

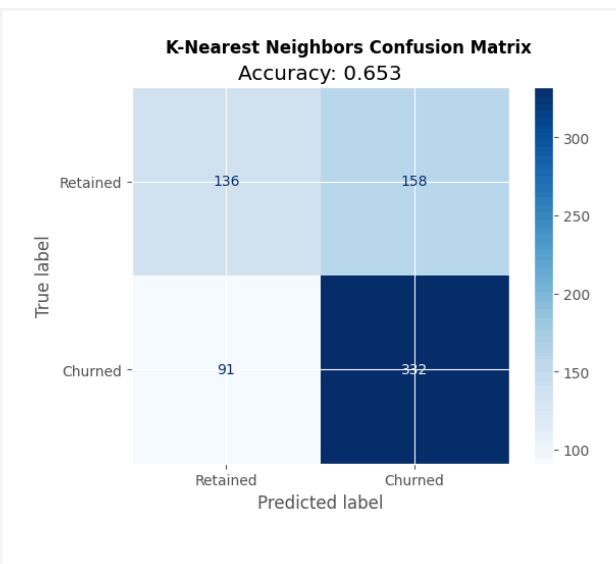
For this model, we used a simple 70-30 train and test split without cross-validation or hyperparameter tuning. As expected, this model produced the weakest performance. Performance metrics on the test set are summarized below.

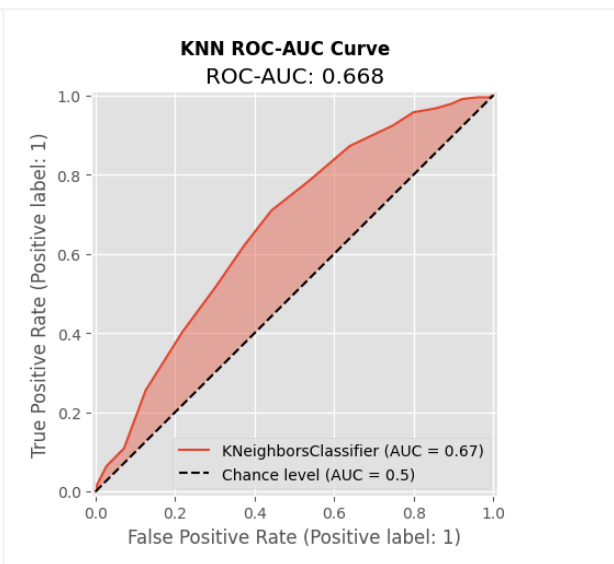
The model struggled particularly with identifying retained customers (recall = 0.46), suggesting that there was limited predicted power when using only four features and no tuning. This provided a clear benchmark for improvement.

KNN Classification Report for Initial Model

	Precision	Recall	F1-Score	Support	Overall AUC
Retained	0.60	0.46	0.52	292	0.668
Churned	0.68	0.78	0.73	423	

KNN Confusion Matrix and ROC-AUC for Initial Model





Expanded Feature Set

To improve predictive performance, we expanded the feature set based on domain intuition about factors likely related to churn. Specifically, we added the following 15 features:

- IsNonAnnual, SingleGradeTripFlag, SchoolType, SchoolSizeIndicator, SchoolGradeTypeHigh, IncomeLevel, FPP, ToGrade, NumberOfMeetingswithParents, Tuition, Days, TotalPax, TotalSchoolEnrollment, FPPtoSchoolenrollment, DepartureDate_month

Because some of these features are categorical, we applied one-hot encoding to convert them to binary variables. Additionally, we revised our splitting strategy to a 70/15/15 (percent) train-validation-test split to allow for more structured hyperparameter tuning and model selection without leaking information from the test set.

Hyperparameter Tuning

We performed hyperparameter tuning using GridSearch with 5-fold cross validation, optimizing over several KNN parameters, including the number of neighbors (K), weighting scheme, and the distance metric power parameter (p). The best performing model configuration used 33 neighbors, distance weights, and a power parameter of 1. The preference for distance weighting indicates that closer neighbors contributed more strongly to classification, while p=1 suggests that Manhattan distance was more effective than Euclidean distance.

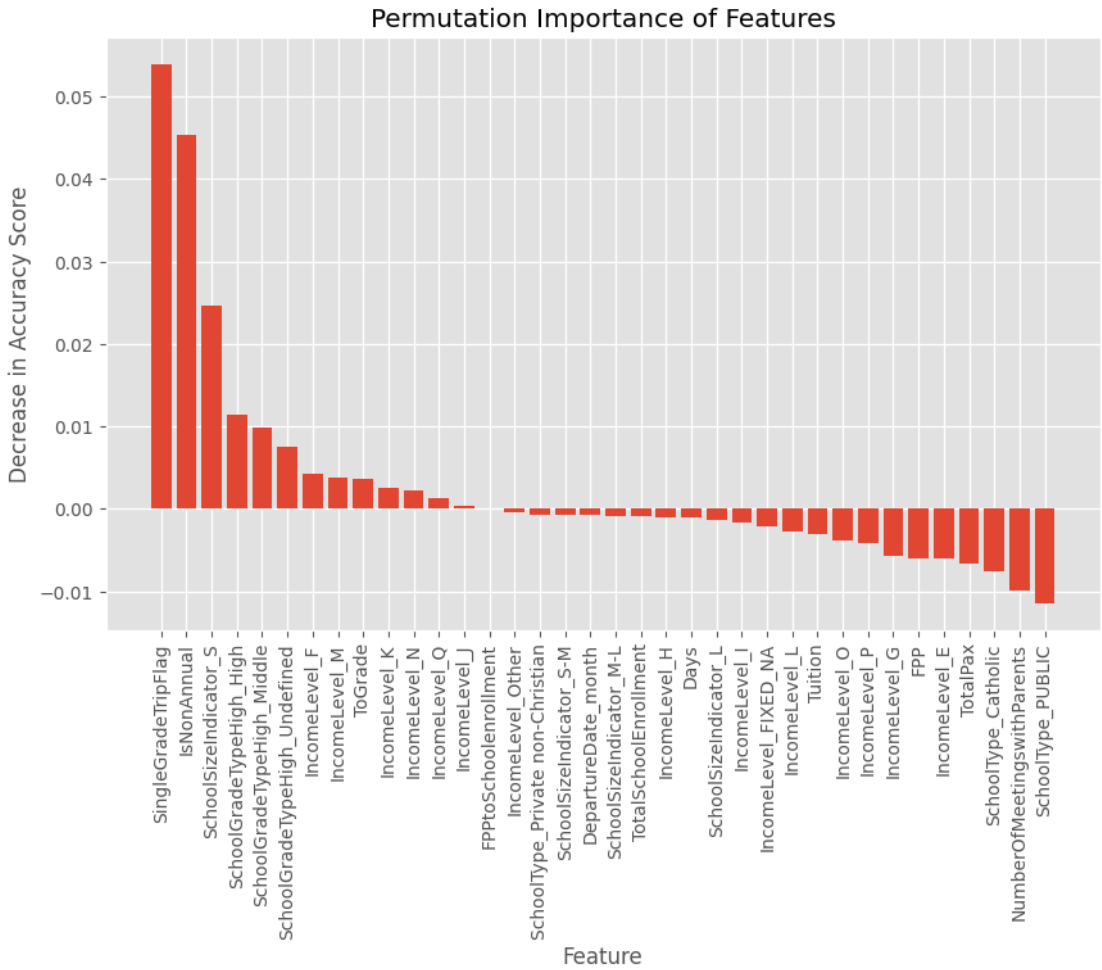
The log of top 10 models as we changed hyperparameters can be viewed below.

n_neighbors	weights	p	CV_Accuracy	CV_Precision	CV_AUC	CV_Recall
33	**distance**	**1**	**0.763157**	**0.756637**	**0.817029**	**0.906621**
43	distance	1	0.761371	0.753029	0.819647	0.910514
29	distance	1	0.76077	0.754935	0.813502	0.904679
31	distance	1	0.76077	0.754943	0.81546	0.904679
32	distance	1	0.760767	0.755341	0.815897	0.903699
42	uniform	1	0.760763	0.752074	0.819872	0.91148
35	distance	1	0.760166	0.753473	0.815322	0.906616
27	distance	1	0.760166	0.755119	0.810492	0.902737
30	distance	1	0.759578	0.753223	0.814814	0.905655
34	distance	1	0.759569	0.753324	0.816274	0.905655

On the validation set, this tuned model achieved an accuracy of 0.758 and an AUC of 0.813. This represents a substantial improvement from the baseline model, indicating that both expanded feature inclusion and systematic tuning enhanced model performance.

Feature Selection through Permutation Importance

To further refine the model, we applied permutation importance to assess the marginal contribution of each feature. Using this method with 30 repeats, we were able to understand how shuffling individual features impacted the accuracy. Features with positive importance values were kept, producing a reduced feature set.



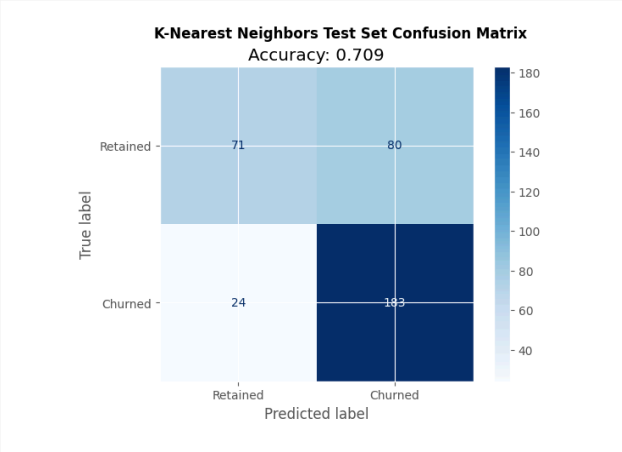
However, after retraining and re-tuning the KNN model on this smaller feature set, performance declined. Specifically, this 3rd model resulted in an accuracy of 0.755 and an AUC of 0.792. This decrease suggests that even features with low singular importance may contribute meaningfully in combination with other features. Accordingly, we continued with the full 15-feature model.

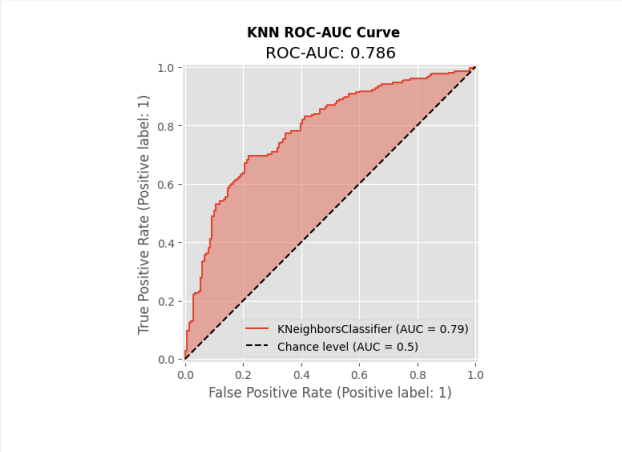
Final Model Performance

Using the tuned KNN model on the full feature set, final performance on the held-out test was:

Accuracy: 0.709 | AUC: 0.786

While test performance was slightly lower than validation performance (suggesting mild overfitting), the final model substantially outperformed the naive baseline. The results indicate that careful feature expansion and hyperparameter tuning meaningfully improved the model's ability to predict churn.





Tree-Based Model (XGBoost)

Feature Selection through L1 Regularization

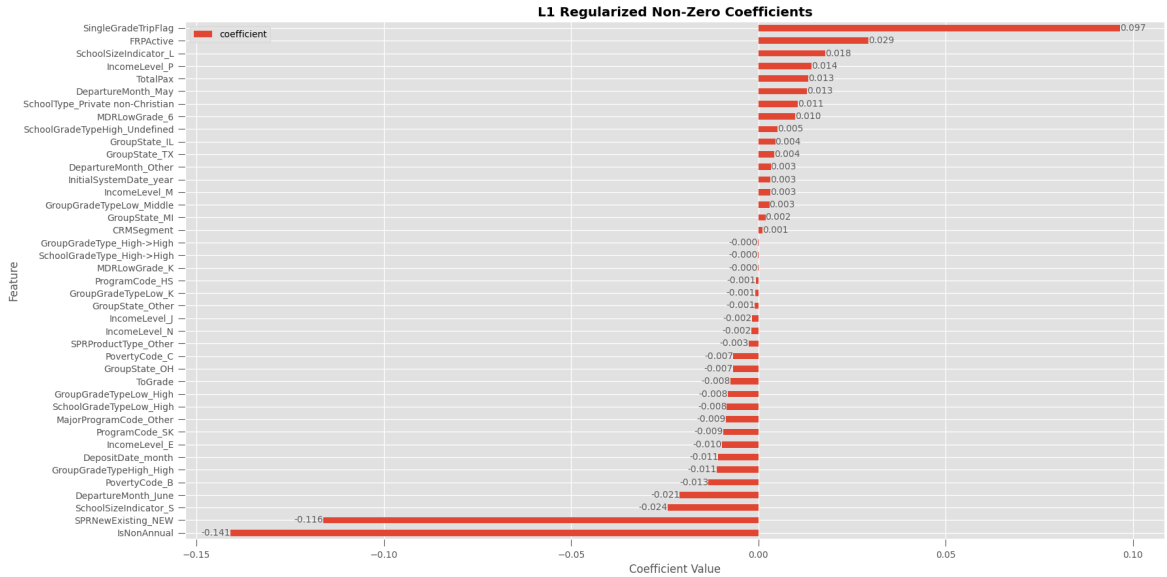
Though XGBoost and other Tree-Based models rarely suffer significant drops in accuracy from highly dimensional training data, careful feature reduction can still deliver benefits. Overt *dependency* on training data dimensions may lead to **overfitting** and poor performance on unseen data– particularly in cases of lossy data collection. Therefore, after our implementation of K-Nearest Neighbors, we opted to use computational regularization methods on our dataset.

L1 and L2 regularization, given a set of estimators, add a **penalty** to the model loss function based on the sum of coefficients used, incentivizing less important features to have diminished coefficient weights. LASSO, or L1 regularization, drives the least important features to exact coefficients of zero (0). This effectively removes unimportant features from the model entirely.

As we did not anticipate *every* feature provided in the Scholastic Travel Company's training data to be useful for determining account attrition *and* sought to reduce dimensionality, we chose LASSO. Using *all* available features from our training set as estimators for the outcome variable, L1 Regularization brought us from **152** (mostly hot-encoded) fields to **41** non-zero coefficients**.**

After Lasso (L1) Regularization

Though 41 fields were important *enough* determiners that L1 Regularization retained them, a few features were distinctively important. Positive values of *SingleGradeTripFlag*, *FRPActive*, *SchoolSizeIndicator_L*, *IncomeLevel_P*, and *TotalPax* made accounts more likely to be retained in descending effect order. *IsNonAnnual, SRPNewExisting_NEW, SchoolSizeIndicator_S, DepartureMonth_June,* and *PovertyCode_B* values made an account more likely to *churn*. No engineered features (i.e. extracted year or month from date fields) were ultimately important.



In the context of the client's needs, the following features are associated with retainment:

- Only a single grade attending the class trip associated with the record
- A higher number of Full Refund Program provisions associated with the trip record
- A larger school size (reflected in SchoolSizeIndicator_L)
- A larger amount of attendees overall (Total Pax)
- A higher income level, but particularly at Income Level P

The following features, contrastingly, are associated with attrition (churn):

- A given record reflecting a *non-annual* trip
- The SRP associated with the given trip being entirely new
- A smaller school size (reflected in SchoolSizeIndicator_S)
- Having 6 to 15.9% of the local area beneath the poverty line, reflected in Poverty Code B
- A trip having its departure date set in June

It is important to note that, while L1 Regularization produces its coefficients for its own estimations, **coefficients were not passed onto XGBoost, only a list of chosen features.**

Hyperparameter Tuning through the Bayesian Method

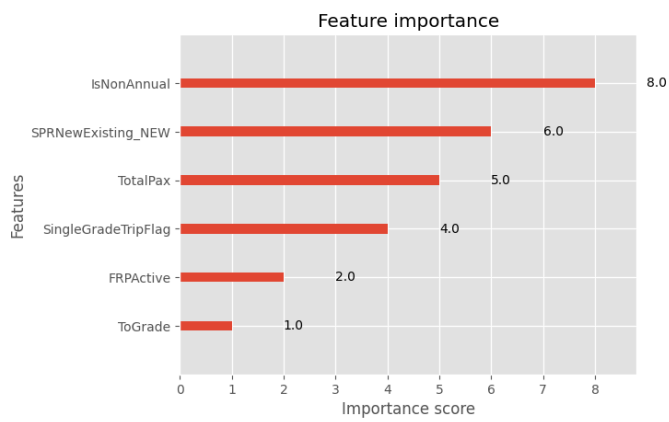
Using the 41 predictors selected through LASSO, we identified hyperparameters for our instance of XGBoost. We opted to use neither GridSearch– iteratively searching over a given "space" of potential parameter combinations– nor RandomSearch– randomly picking sets of options from this parameter space– but instead used Bayesian optimization.

Bayesian optimization, in essence, uses a probabilistic approach to selectively search through a parameter space: how likely is an improvement in the model given a particular alteration? Following that logic, parameters are tweaked sequentially over a user-provided amount of instances. The hyperparameters that, over these instances, produce the least loss against a "validation" set of data (and thereby are the most accurate in their predictions) are returned as hyperparameters. Included are the most effective combinations found across 100 iterations.

Top 10 Most Effective Hyperparameters (with #1 as best combination)

#	accuracy	max_depth	gamma	reg_alpha	reg_lambda	colsample_bytree	min_child_weight	n_estimators
1	0.8161559888579	14	4.182230185772	42	0.14076472040	0.9600498229607	1	51
2	0.8161559888579	17	1.1335883495504	43	0.99806911672	0.9964066317290	4	88
3	0.8161559888579	16	3.2312552883149	51	0.13865448852	0.966892	4	61
4	0.8161559888579	16	3.2025255881851	48	0.12546027348	0.9652250269690	4	44
5	0.8133704735376	18	7.587456102863	50	0.14696278728	0.9455746297888	2	23
6	0.8133704735376	12	4.228556	54	0.50290375466	0.7511896404875	3	6
7	0.8133704735376	15	1.2615349123724	41	0.98540241496	0.985119	5	70
8	0.8133704735376	8	1.6002597215419	40	0.620084	0.9271542219799	1	88
9	0.8133704735376	9	2.2529146868855	67	0.533007	0.6823076695140	4	49
10	0.8133704735376	10	5.767444723588	54	0.74778672329	0.9579888727414	3	62

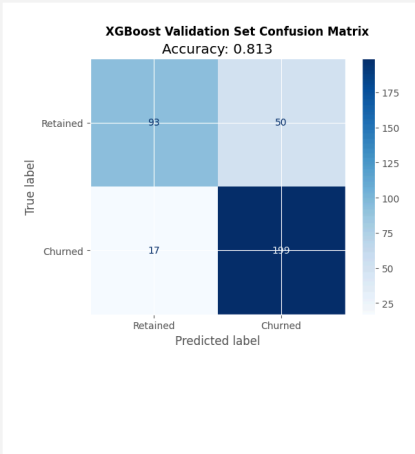
Performance and Emergent Features of Importance

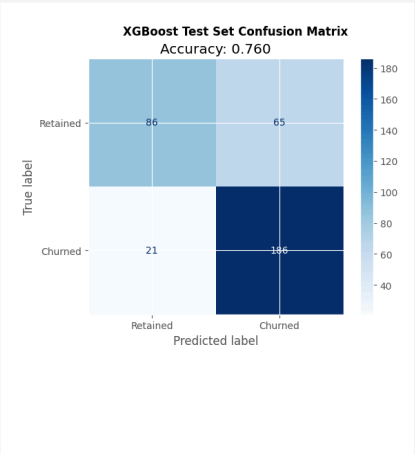


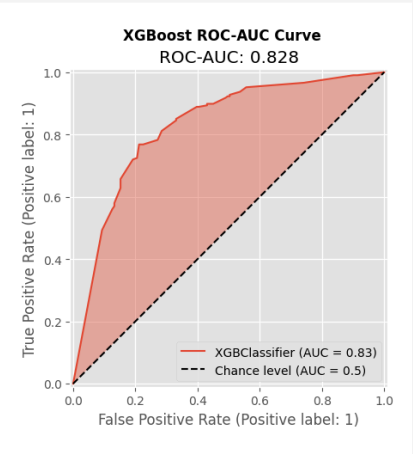
The resulting hyperparameters and selected features were then used to train XGBoost– and, in succession, used to test its predictive power against a Validation Set and Test Set of data. As expected (given that loss against the Validation Set was minimized through the Bayesian Method), performance is particularly strong on the Validation Set, yet weaker on the Test Set. The model overall, however, possesses a strong ROC-AUC score at 0.828.

Ultimately, the most important features for *this* model became: *ISNonAnnual*, *SPRNewExisting_NEW*, *TotalPax*, *SingleGradeTripFlag*, *FRPActive*, and *ToGrade*.

METRICS (Weighted AVG): Accuracy: 0.76 | Precision: 0.77 | Recall: 0.76







Alternative Model (Support Vector Machine)

Selection and Basics

We chose [Support Vector Machines](#) for our alternative model. Support vector machines (SVM) are a set of methods that can be used in a variety of supervised machine learning applications, but we use it for classification. At a high-level, the model maps a hyperplane onto the feature space. The model attempts to place this hyperplane in a way that attempts to cleanly place each class on the opposite side of the hyperplane while maximizing the distance between the hyperlane and the nearest point in each class from the training set. These nearest data points to the decision boundary are the "support vectors." Non-linear kernel extensions of this process map vectors onto a higher-dimensional space in order to better account for data point spacing across the classes.

Set-Up and Hyperparameters

We implemented the support vector machine approach through the svc function from scikit learn's svm sub-module. Building on the previous model work, we used the features provided by the L1 regularization process. Outside of calling the initial function for use in the estimator, we needed to identify the key hyperparameters that the model uses to set-up the grid search for hyper parameter tuning.

The first-order decision is to determine the kernel type. Given our understanding of the data, linear and RBF seemed the most appropriate to test. Linear and radial basis functions are the most commonly used kernel choices for this methodology, with RBF being the common starter kernel for various use-cases due to its ability to handle the data when linear bisection of the data is less obvious. We computed both of these options in our grid search.

In addition to the kernel choice, each has a distinct combination of hyperparameters to tune. While the linear kernel only needs the C parameter, the RBF kernel also needs a gamma parameter. They can be defined as:

- C: a penalty term for misclassification. A higher C value punishes misclassification more, creating a more jagged and bespoke decision boundary. A lower C value is more accepting of misclassification and creates a smoother decision boundary. A lower C risks higher bias; a higher C risks variance.
- Gamma: a term for how much influence a given data point has. A lower gamma means the decision boundary cares more about farther away points, and a higher gamma cares less about those points. A lower gamma risks higher bias; higher gamma risks variance.

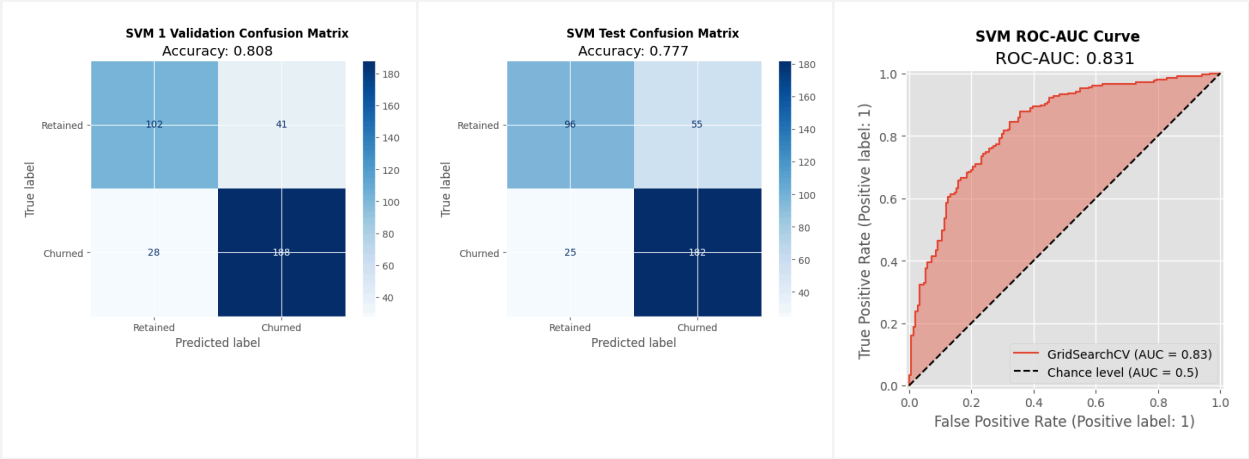
To find the optimal kernel, C, and, if necessary, gamma, we implemented grid search to find the optimal values. The scikit learn package advised to space out each value of C and gamma by orders of magnitude (0.1, 0.01, 0.001, etc) and let the grid search determine the general points to anchor later tuning on. Notably, the linear model took dramatically longer time to run at higher levels of C, leading us to lower the upper limit for the linear kernel for practical implementation. The result: C: 1, gamma: 0.01, kernel: rbf.

The next step was to zoom in on these boundaries. Given how far each interval in the grid search was spaced, it seemed plausible the optimal solution would be on either side of each numeric value, so we created a list of much closer values on either side of the C of 1 and gamma of 0.01. We anchored to RBF based on the previous tuning results. Interestingly, the grid search found a new set of parameters that kept C at 1 but moved gamma 0.005; the result was a marginal decrease in accuracy on the validation set. Given this decrease in performance, we concluded that the initial parameters from step 1 were the better option on raw accuracy performance.

After hyperparameter tuning, we used the test set to evaluate the final model, with results below.

Evaluation

METRICS (Weighted Average): Accuracy: 0.78 | Precision: 0.78 | Recall: 0.78



Alternative Model Strengths and Weaknesses

Support vector machines are an interesting model to apply to this use-case with a few strengths and weaknesses. The major strength of SVM is that it is exceptional in high dimensional spaces. We used only around 40 features to construct this model and had more than enough rows to meet common industry thresholds of feature count times ten (400 rows). That meant we could feel exceptionally confident this model was capable of handling our dataset's size.

A major challenge of using this model was not about performance but rather interpretability and the ability to use similar feature importance methodologies available in, for example, XGboost. There are no native probability outputs for classification problems in either the linear or RBF kernel models. This means it is more difficult to zoom in on how likely someone is to churn when we make later recommendations based on our identified "best features." While alternative conversion processes exist, they add an extra layer of interpretation difficulty. Additionally, since we used a non-linear kernel in the final model, we lost the ability

to use direct feature importance specific to the inner workings of the model and instead have to use permutation importance which added significant run-time.

Model Comparisons and Selection

Model	*Accuracy*	*Precision (W. AVG)*	*Recall (W. AVG)*	*ROC-AUC Score*
K-Nearest Neighbors	0.71	0.72	0.71	0.79
XGBoost	0.76	0.77	0.76	0.828
SVM (Final Choice)	0.78	0.79	0.78	0.831

K-Nearest Neighbors

KNN was computationally efficient and easy to configure. At the same time– as expected from application to a largely categorical dataset– KNN lagged behind other models when attempting to maximize accuracy. KNN also had *alarmingly* low negative predictive when identifying churn, with the corresponding recall at **0.47**, bordering on random. Improperly responding to churned clients (whom provide $1,604,625 of profits in the training set) leads to reduced ROI. False positives, whether in the case of retention or churn, lead to wasteful allocation of resources.

XGBoost (Tree-Based Model)

XGBoost served as a fairly robust and powerful predictor. Given the high degree of categorical information included with STC's surveys, XGBoost, as a tree-based model, performed notably better than KNN out-of-the-box. Not only that, but XGBoost was particularly swift to implement and even set hyperparameters for. Unfortunately, its *negative class* recall suffered when applied to test set data, bordering on random at **0.57**– false negatives (meaning unidentified churn) are undesirably frequent under the XGBoost model, making it unreliable for targeting this segment.

Support Vector Machines (Alternative Model)

A Support Vector Machine (SVM) was highly effective for classifying our customer data, despite its highly categorical nature. Moreover, SVM efficacy endured even when applied to the test set, only seeing a 0.031 reduction in accuracy between the validation and test sets. Compared to XGBoost, however, SVM was slightly more computationally intensive. Though partially attributable to the use of a grid-based search (compared to XGBoost's probabilistic implementation), SVM's inherent use of *quadratic* optimization makes it particularly intense when trained on larger datasets. At the same time, SVM delivered a better negative class recall at **0.64** and achieved overall better weighted performance. Thus, it was selected for prediction.

Recommendations

At a high level, we recommend that STC align greater resources towards client churn risk. This not only includes deploying more nuanced marketing campaigns, but reacting to how segments might be better served through targeted sales or account management. We treat STC's services as B2B (business-to-business) with "teacher-organizer" as primary Point-of-Contact, though acknowledge that institutional budgets and higher-level stakeholders may constrain actual customer behavior. We find that high-risk segments are associated with new trips or non-annual / off-year trips, and low-risk segments are more likely to be single-grade trips, include relatively high users of the Full Refund Program, or possess larger total attendee sizes. This aligns not only with our exploratory K Nearest-Neighbors findings, but the determinations of L1 Regularization, XGBoost feature importance ranking, and SVM feature permutation analysis.

1. Segment Trips by Annual Status (Highest Priority)

STC should develop and deploy a two-tier engagement model based on school trip cadence.
Moreover, it should redefine a "retained customer" in light of known non-annual patterns of service and engagement with particular teacher-organizers. We term these as "off-year" clients.
As STC presently defines a client as "retained" if and only if they engage in the same product *the very next sequential year*, it synonymizes consistent (albeit irregular) customers with total one-time users. As "churned" clients represented over $1.6M in known profits, this obscures any actual revenue-making distinctions. We suggest the following for "annual" and "off-year" clients:

- **Annual Clients**
    - *Reward customer loyalty*: incentivize continual account renewal with promotions, targeted messaging, and discounts for multi-year packages
    - *Establish high-touch communication from account managers*: complaints, concerns, and questions should be addressed within a more immediate timeline
    - *Foster a sense of community*: personalize messaging and outreach towards institutional procurers and leaders, creating a sense of "care" from STC
    - *"Ramp up" follow-up marketing after trips*: follow-up with clients as soon as possible after service in order to retain them within the annual segment
- **Off-year Clients**
    - *Shift to leaner, lower-cost marketing segment*: deprioritize off-year clients during "lull" and seek to "maintain" rather than "reward" teacher-organizer relationship
    - *Target with light-touch, periodic communication*: In the interest of maintaining client, send organizers cyclical reminder (e.g. postcards or emails) for renewal
    - *Nudge toward "annual" behavior conversion*: Advertise distinct annual service benefits, inducing "Fear of Missing Out" for institutional decision makers
    - *Divide into sub-segments to optimize sales resource allocation*: Using other indicators for retainment as context, prioritize more "ideal" customers, such as higher income institutions or institutions more likely to send larger trip groups

In following these recommendations, STC also responds to two distinct churn windows that better reflect the realities of its consumer base: a "short churn," in which a client does not engage for the next immediate year, and a "long churn," where a client is lost indefinitely. Though "short churn" clients do not promise immediate gains, light-touch engagement helps maintain a *consistent*, multi-cycle relationship. Investments in "long churn" clients can ultimately be wasteful, meaning that marketing campaigns would struggle to produce ROI on this segment.

2. "Level Up" Retention Efforts for New Trips (Secondary priority)

In partnership with Sales leadership, the STC Marketing Team should develop and roll out a first-year retention strategy for newly acquired customers. This may include the following:

- *Target with "highest"-touch communications:* Consider new clients as "highest risk" for "long churn" and therefore place within "highest priority", multi-channel marketing stream
- *Provide with dedicated and tailored account management*: Within Sales, develop expertise and capacity for handling uniquely "new" client concerns and pain points
- *Catalogue client decision-makers and procurers as contacts*: Recognizing client as B2B engagement, build deep relationship with persons *beyond* initial teacher-organizer unit
- *Expand, deepen, and diversify relationship as fast as possible*: Encourage clients to consider not only "annual" conversion, but complimentary trip products and services
- *Gather "Voice of Customer" as early as possible*: After completion of first client engagement (i.e. trip completion), rapidly gather feedback from *all* involved parties

By ensuring that new clients (including institutional procurers, teachers, and even parents) feel welcome and supported by STC, Marketing and Sales can reduce churn and increase loyalty.

3. Deprioritize Single-Grade Trips in Retainment Marketing (Tertiary)

STC should redirect its marketing campaigns away from single-grade trips and toward higher-risk customer segments. Rather than diminishing attention altogether, single-grade trip clients should be served by Sales account managers positioned to preserve relationships. Thus:

- *Only incentivize Single-Grade Trips for Prospects (i.e. New) Clients:* Single-grade trips already highly tend towards renewal. Therefore, advertise them to *potential* clients.
- *Fold "Single-Grade" into "maintenance" strategy*: After a single-grade teacher-organizer moves out of "new" status after confirmed renewal, hand off attention to Sales specialist

4. Strategically emphasize operational capacity for large attendance trips (Tertiary)

Within our analysis, we observe that larger trips (i.e. those with a higher TotalPax) more often tend towards being retained for next-year rebooking. While this correlation may well be totally non-causal, it signals an untapped market intelligence space for product refinement. Within the STC Marketing team, the following can be overtaken before and during the spring sales season:

- *Highlight long-term, high TotalPax client relationships in advertising*: Use collateral from ongoing, annual engagements (trips) to showcase success for prospective clients
- *Provide B2B referral opportunities, as to benefit from word-of-mouth*: Collect and actively reward individual-level feedback from high TotalPax engagements for brand awareness
- *Collect specific, employee experience (EX) from high TotalPax program managers:* Direct service staff input from this product can provide "what works" intelligence

Appendix

Data Dictionary

Data Field Name	Example	Description
ID	1	Self-explanatory.
Program.Code	HD	This is a very granular code that describes where the trip went and what it did. HN, for instance, is a history program that runs in New York.
From.Grade	8	This is the lowest grade in school of a participant on that program.
To.Grade	8	This is the highest grade in school of a participant on that program.
Group.State	IN	This is the two-letter designator for the state in which the originating school is located. OTHER stands for rare geographies that appear in the data only once.
Is.Non.Annual.	1	1/0 indicating if the group from this school typically skips a year in between programs. These will rarely repeat the very next year.
Days	3	The number of days the group was on the program and with one of the instructors.
Travel.Type	A	Mode of travel from the originating school location to the starting location of the program (A = Air, B = Bus, T = Train).
Departure.Date	2/19/2011	The date that the group left its originating school.
Return.Date	2/21/2011	The date the group returned to its originating school.
Deposit.Date	10/20/2010	The date by which registrants are supposed to have at least an initial deposit prior to departure. The time in the school year when certain events occur can be important; for instance, there are no deposit dates in the summer since no one would be around to act on them.
Special.Pay	NA	The most important of these are school accounts (SA). That means that, contrary to the usual practice, the teacher collects all of the money and then remits it in bulk to STC. The normal arrangement is STC handling all of the cash collection from parents/students.

Data Field Name	Example	Description
Tuition	1174	This is the price it costs each full-paying participant (FPP) to go on the program. West-coast air trips are more expensive per person than midwestern bus groups.
FRP.Active	72	FRP is the full refund program. This is the number of FPPs on the trip who bought trip-cancellation insurance.
FRP.Cancelled	13	This is the number of FPPs on the trip who bought trip-cancellation insurance, but then cancelled it.
FRP.Take.up.percent.	0.6857	This is the percentage of the FPPs who bought the FRP and ended up paying for it.
Early.RPL	3/2/2010	This is the date that the first communication went out to the group. Often this can be 12 to 18 months before the trip actually departed.
Latest.RPL	8/10/2010	This is the date that the last communication inviting people to join the group went out. Often this can be 6 to 9 months before the trip actually departed.
Cancelled.Pax	15	This is the number of passengers who signed up with a $100 deposit but then cancelled before the group departed.
Total.Discount.Pax	7	This is the total number of extra passengers who went along without paying full price (or typically anything). These would be the chaperones and the teachers.
Initial.System.Date	3/2/2010	This is the date when the teacher first agreed to get this trip organized. It is typically the earliest of the dates relative to group activities.
Poverty.Code	A	Poverty code for the area in which the originating school (and by extension, most of the parents who will be paying for the trip) resides based on estimated percentage below the poverty line. A is 0 to 5.9, B is 6 to 15.9, C is 16 to 30.9, D is 31 or more, E is unclassified, Space if DISTCLASS = U (Supervisory Union).
Region	Other	This is a larger aggregation of state areas. Some large states, like California, are their own region. Others are combined.

Data Field Name	Example	Description
CRM.Segment	1	This is a type of school code used in the customer-relationship-management (CRM) system to describe the school. The codes are numbered 1–11 but are in no particular order; proprietary, but it is a designation of a customer type that may be helpful.
School.Type	PUBLIC	Public or private.
Parent.Meeting.Flag	1	1/0 indicating whether a parent meeting was held. These are typically strong indicators of parent engagement and of a teacher who understands that these can be important to successfully organizing one of these out-of-school programs.
MDR.Low.Grade	7	This is the lowest grade in the originating school.
MDR.High.Grade	8	This is the highest grade in the originating school.
Total.School.Enrollment	955	This is the total enrollment of the school (to differentiate big schools from little ones).
Income.Level	P	Like poverty code, an indication of ability of parents to pay for these programs. A is lowest, Q is highest, Z is unclassified.
EZ.Pay.Take.Up.Rate	0.2286	This is a % of the FPPs that sign up for an automatic bank draft installment plan.
School.Sponsor	0	This is an indication (1/0) of whether or not the school is officially sponsoring the trip. Mostly, though these programs draw from the same school, they are typically run independently.
SPR.Product.Type	East Coast	A high level of aggregation of the very granular tour types.
SPR.New.Existing	EXISTING	EXISTING means that the group has traveled with STC before—most often the year before. NEW, with few exceptions, means that the school has never traveled before with STC.
FPP	105	This is the actual number of FPPs who went on the trip.
Total.Pax	112	This is the actual number of total passengers (including chaperones and teachers) who went on the trip.
SPR.Group.Revenue	125735.4	This is the total amount paid for all of the participants to go on the program from that group.

Data Field Name	Example	Description
NumberOfMeetingswithParents	0	Number of meetings with parents prior to the trip.
FirstMeeting	11/18/2010	The date of the first meeting with parents (NA if none held).
LastMeeting	11/28/2010	The date of the last meeting with parents (NA if none held, may be same as the first meeting if only one meeting was held).
DifferenceTraveltoFirstMeeting	93	The number of days from the first parent meeting to travel date.
DifferenceTraveltoLastMeeting	103	The number of days from the last parent meeting to travel date.
SchoolGradeTypeLow	Elementary	The lowest grade type in the school.
SchoolGradeTypeHigh	Elementary	The highest grade type in the school.
SchoolGradeType	Elementary->Elementary	Combination of the above denoting the type of school.
DepartureMonth	January	Month of departure.
GroupGradeTypeLow	K	The lowest grade type in the group that travels.
GroupGradeTypeHigh	Elementary	The highest grade type in the group that travels.
GroupGradeType	K->Elementary	Combination of the above denoting the type of the group that travels.
MajorProgramCode	H	Aggregation of the granular program code; the first letter of the program code.
SingleGradeTripFlag	1	Indicator for the trip taken by a group comprising students from the same grade.
FPP.to.School.enrollment	0.06364617	The ratio of FPP to school enrollment.
FPP.to.PAX	0.936507937	The ratio of FPP to total PAX on the trip.
Num.of.Non_FPP.PAX	4	The number of PAX who are not FPP.
SchoolSizeIndicator	L	A label for the size of the school (S, M, L, S-M, M-L), by quintiles of sizes.